



**Bicol University Polangui, Albay**  
**Web System and Technology**  
**2nd Semester S.Y 2025-2026**



**IT 112 WEB SYSTEMS AND TECHNOLOGIES**

# **BU TRANSAKTO**

**PHSE 7 LAB REPORT**

**GROUP NAME:** KMJ's

**MEMBERS:**

BELLEN, JUVY

HERMO, MA. MERA

INTIA, KIMBERLY ANNE

REALUBIT, JANET

SEPTO, JESSECA

**COURSE & BLOCK:**

BSIT-2A

## **Title:** BU TRANSAKTO- Registrar System

**1. System Overview & Objective:** BU Transakto is a full-stack web application designed to digitalise and streamline document requests at Bicol University – Polangui Campus. Students can request transcripts, certificates, and other documents online, track their request status in real time, and receive notifications. Staff members manage all requests, update statuses, and view analytics. The system also functions as a Progressive Web App (PWA), making it installable on devices and capable of offline caching for static assets.

### **Key Goals**

- Eliminate paper-based request forms.
- Provide a transparent, real-time status tracking for students.
- Reduce staff workload by centralising request management.
- Ensure the application is responsive, secure, and works offline as a PWA.

## **2. Features Implemented**

### **Students:**

- Registration & Login (with role-based redirection)
- Document Request Form (dropdowns for document type, purpose, course, year, semester, number of copies)
- Edit / Delete own pending requests only
- View request status and transaction history
- Notifications (real-time when status changes)
- Feedback (optional anonymous submission)
- Profile Management (edit all fields, email must contain @, all fields required)
- Change Password (minimum 6 characters)
- Dark Mode Toggle

**Staff:**

- Dashboard with statistics: total requests, pending, processing, completed, rejected, total students
- View all student requests (with search & status filter)
- Update request status (pending → processing → completed / rejected)
- View registered students and their request counts
- Read student feedback (names anonymised when requested)
- Change Password General
- Full CRUD for requests (Create, Read, Update, Delete) with proper permissions
- JWT Authentication + role-based access control
- Responsive UI (mobile, tablet, desktop)

Progressive Web App - Web App Manifest (manifest.json) - Service Worker (sw.js) caching static assets - Offline fallback to index.html - Installable on devices (Add to Home Screen)

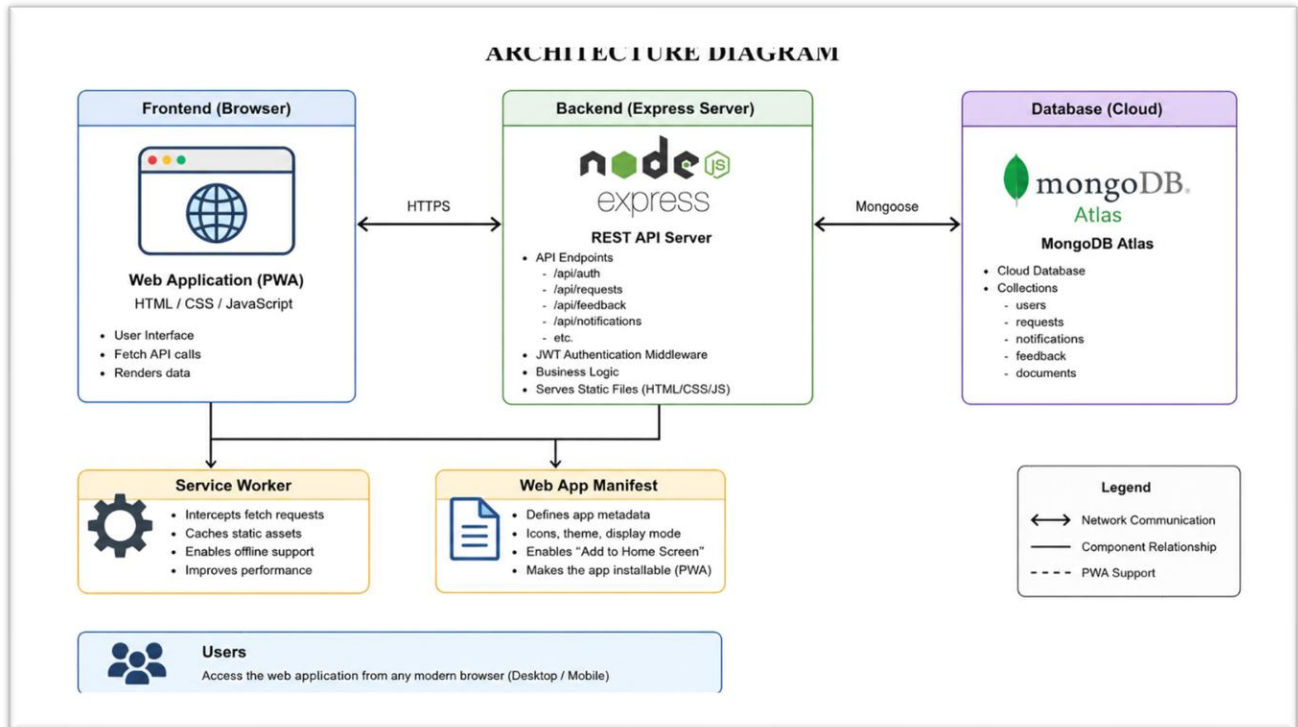
- Toast notifications for all user actions

**General**

- Full CRUD for requests (Create, Read, Update, Delete) with proper permissions
- JWT Authentication + role-based access control
- Responsive UI (mobile, tablet, desktop)
- Progressive Web App
- Web App Manifest ( `manifest.json` )
- Service Worker ( `sw.js` ) caching static assets
- Offline fallback to `index.html`

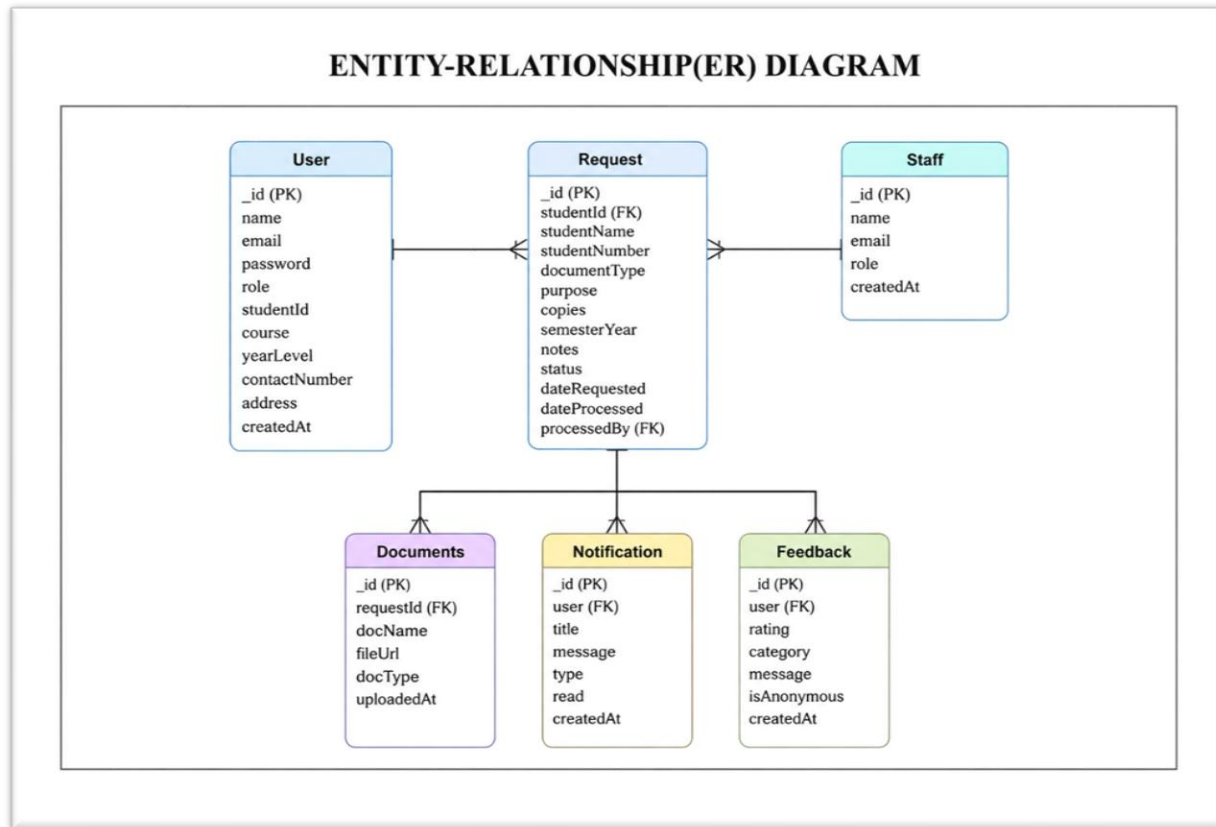
- Installable on devices (Add to Home Screen)
- Toast notifications for all user actions

### 3. Architecture Diagram



The Architecture Diagram shows how the system components work together. The Frontend (HTML, CSS, JavaScript) is accessed through a browser and communicates with the Backend (Express.js server) using HTTPS. The backend handles API requests, authentication, and business logic, then connects to MongoDB Atlas for data storage. The system also includes PWA features such as a Service Worker for caching and offline support, and a Web App Manifest for app installation.

## 4. ERD Diagram



The ER Diagram shows the database structure of the document request management system. It includes the main entities: User, Request, Staff, Documents, Notification, and Feedback. Users can create requests, receive notifications, and submit feedback. Staff members process requests, while documents store request-related files. The diagram also shows the relationships between these entities for organized data management.

## 5. GitHub Repository & Commit Logs

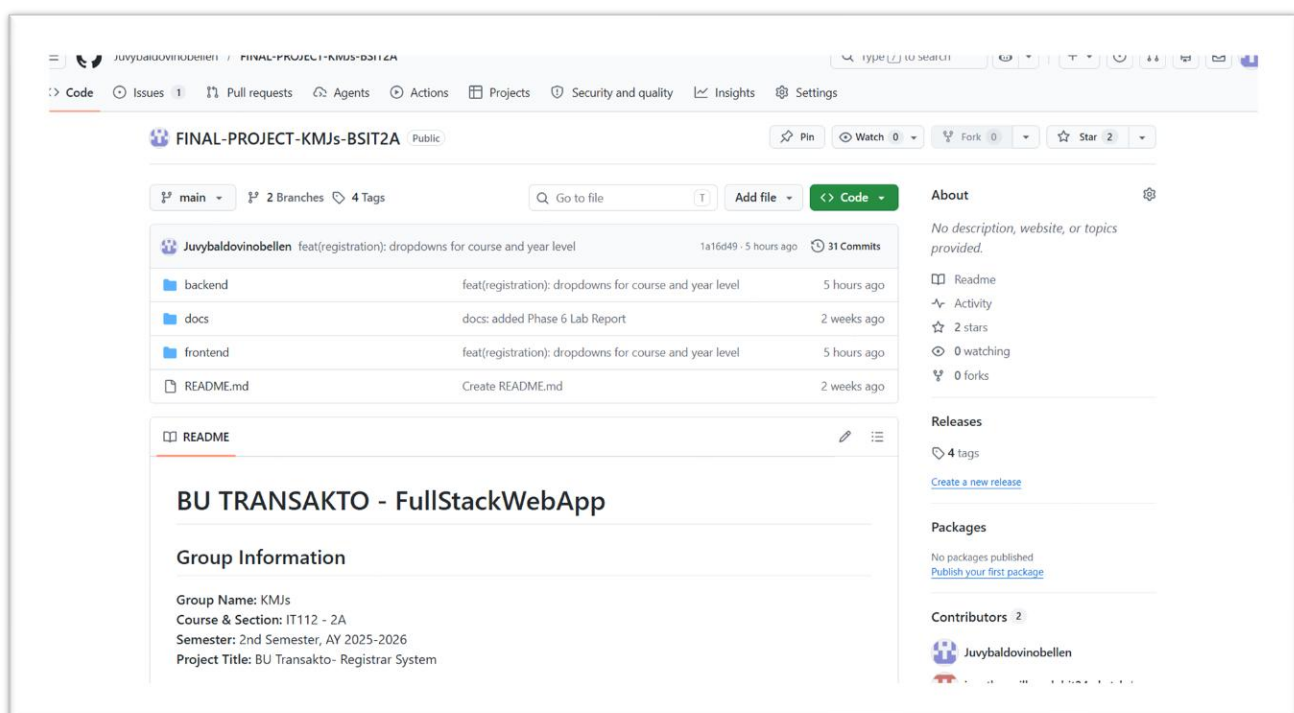
Repository URL:

<https://github.com/Juvybaldovinobellen/FINAL-PROJECT-KMJs-BSIT2A.git>

## Commit History

### Latest Commits (Newest to Oldest)

1. 1a16d49 (v1.0-final) – feat(registration): dropdowns for course and year level
2. 0f5b947 – fix(feedback): enforce authentication and anonymous notification
3. e9e817e – Added changes in Edit Request
4. af93c89 – feat(student): enhance edit request modal with dropdowns & required fields
5. a1bd33b – Added dropdown for Semester/Year
6. 28d2556 – Update notification system and UI improvements
7. e9a879d – fix: use relative API path for production (remove hardcoded localhost)
8. 623756a (week15-pwa) – feat(pwa): implement Progressive Web App (manifest, service worker, offline caching)
9. f422fe5 – docs: added Phase 6 Lab Report
10. 79009dc – fix(staff): add sidebar logout button functionality



## 6. Render Deployment & Live URL

<https://bu-transakto.onrender.com/>

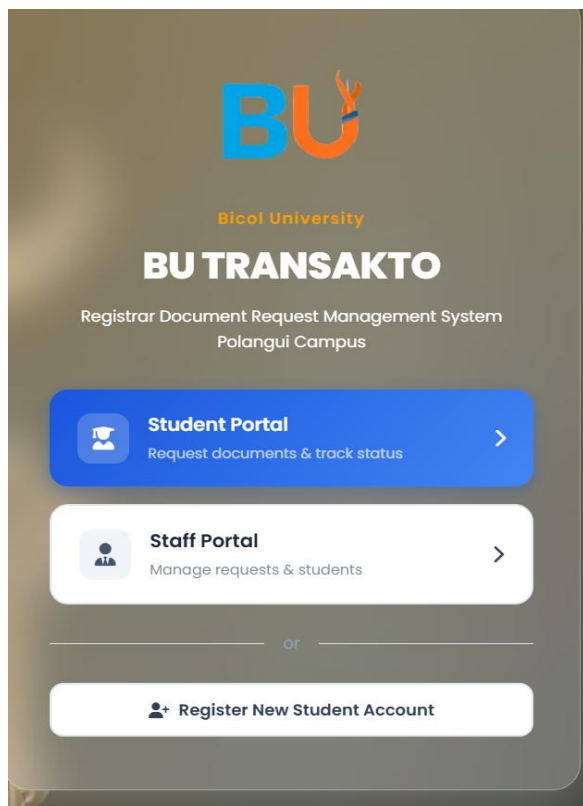
Deployment details:

- Platform: Render (Web Service)
- Root directory: . (project root)
- Build command: `cd backend && npm install`
- Start command: `node backend/server.js`
- Environment variables:  
MONGO\_URI=mongodb+srv://KMJs:KMJs@finalprojectbutransakto.ftmikhy.mongodb.net/?appName=FinalProjectBUTransakto  
PORT=5000

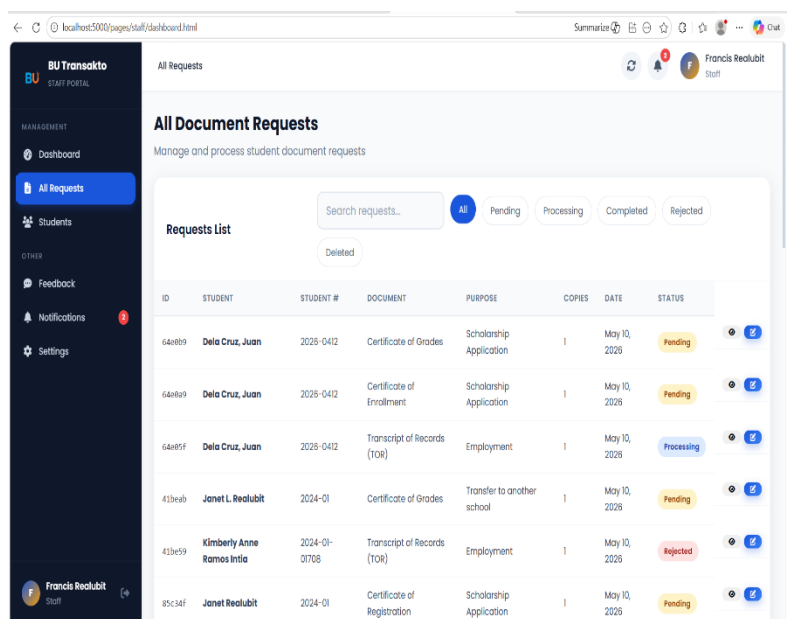
JWT\_SECRET=aefb48916ad1a3291bee3e7f33cfb82afe3ec70b0b1baba8eaa343a1919f3642

### LIVE SITE

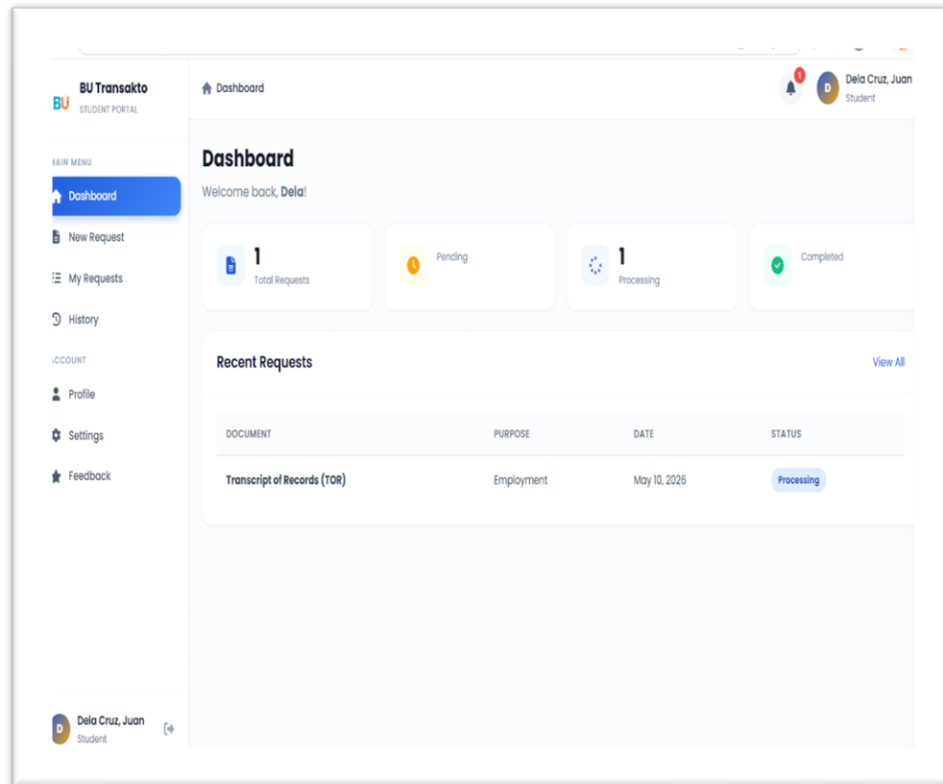
Landing Page



Staff Dashboard



## Student Dashboard



## 7. Summary of Division of Tasks

### Backend Developer (Janet L. Realubit)

As the Backend Developer and Project Lead, I was responsible for ensuring that all endpoints and system functions worked properly while also coordinating the team's progress. I handled testing for invalid inputs to maintain the stability and reliability of the application. At the same time, I monitored pending tasks and made sure the project was prepared for the final demonstration. This role helped me improve my problem-solving, leadership, and communication skills. It also taught me the importance of teamwork and proper coordination in completing a successful project.

### Frontend Developer (Kimberly Anne Intia)

As the Frontend Developer, I focused on improving the design and functionality of the user interface. I worked on making the layout responsive and accessible so users could easily navigate the system on different devices. I also fixed bugs



and improved the overall appearance of the application. Through this role, I learned the importance of creating a smooth and user-friendly experience.

### **Database Manager (Ma. Mera Hermo)**

As the Database Manager, I was responsible for maintaining data consistency and organizing the database structure. I checked if the schema matched the requirements of the system to avoid data issues. This role helped me understand the importance of proper data management and accuracy. I also learned how databases support the overall performance of an application

### **GitHub Manager (Juvy Bellen)**

As the GitHub Manager, I handled merging branches and organizing the project repository. I also cleaned up unnecessary files and updated the README to make the project more presentable and understandable. This role improved my knowledge of version control and collaboration tools. It also taught me the value of maintaining an organized workflow in team projects

### **Documentation Officer (Jesseca Septo)**

I prepared user guides, technical documentation, diagrams, and testing reports for the project. My role was important in making sure that all project details were clearly explained and easy to follow. I learned how proper documentation helps users and developers better understand the system. This experience also improved my writing and organizational skills.